

Web Search (2): Sponsored Search, SEO, Duplicate Detection, and Crawling

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

early keyword engines and scalability limits → sponsored search monetises ranking attention → Google separates algorithmic results from ads → user intent shapes result type → SEO exploits ranking signals → duplicate detection protects index quality → shingles and MinHash approximate near-duplicate similarity → crawler starts from seeds and expands frontier → politeness/robustness/freshness constrain crawling architecture

11/5/25

Brief History

- Early keyword-based engines (1995-1997)
 - Altavista, Excite, Infoseek, Lycos, AOL
 - Traditional IR techniques
 - Scalability is an issue
- Paid search ranking: Goto (morphed into Overture.com → Yahoo!)
 - Your search ranking depended on how much you paid
 - Auction for keywords
 - Called “sponsored search”
 - CPC (Cost Per Click)
 - CPM (Cost Per Thousand Impressions)

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3

CPC / CPM / RPM

- With new services on the web → RPM
- RPM: Revenue per 1000 video views
- Read more:
 - Understand ad revenue analytics
 - <https://support.google.com/youtube/answer/9314357>

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4

2

这节课一句话

这节课一句话：Web search 从历史上看是“自然排序、广告商业化、SEO 对抗、去重和爬虫工程”共同演化出来的系统，而不是一个单纯的 ranking formula。

1. 1. Web search history: 先有关键词，再有广告，再有图链接

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4

2

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人话直觉

早期搜索像在超大文件夹里 Ctrl+F；后来有人发现，搜索结果位置本身就是黄金广告位。

精确定义 / 机制： 1995-1997 年的 Altavista、Excite、Infoseek、Lycos、AOL 等主要使用 keyword-based traditional IR techniques，问题是 scalability 和 ranking quality。Goto/Overture 引入 paid search ranking：关键词拍卖，排名与支付相关，形成 sponsored search。CPC 是 cost per click，CPM 是 cost per thousand impressions；视频等新服务中还会用 RPM，即 revenue per 1000 video views。

考试问法： 历史题按时间线答：early keyword engines → paid ranking/sponsored search → Google link-based ranking。商业指标题要区分 CPC/CPM/RPM。

2. 2. Algorithmic results vs sponsored ads: 自然排序和广告别混为一谈

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Brief (non-technical) History

- 1998+: Link-based ranking pioneered by Google
 - Blew away all early engines
 - Great user experience in search of a business model
 - Meanwhile Goto/Overture's annual revenues: ~ \$1 billion
- Result: Google added paid search "ads" to the side, independent of search results
 - Yahoo followed, acquiring Overture (for paid placement) and Inktomi (for search)
- 2005+: Google gains search share, dominating in Europe and very strong in North America
 - 2009: Yahoo! and Microsoft combined paid search offering
- 2024+: AI + RAG systems (*no clear winner yet!*)

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Brief (non-technical) History

SEARCH ENGINE WARS 1993-2021

Logos at the bottom: DuckDuckGo, Baidu, Aol., Bing, Yahoo!, Ask.

Source: Wotif Magdy, TTDS 2025/2026

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3

原 PPT 第 3 页

人话直觉

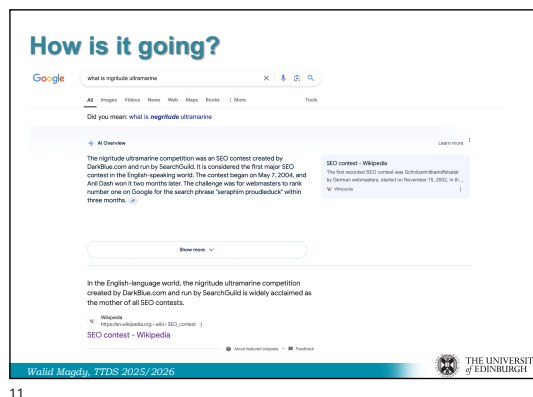
搜索页面上有两条河：一条是算法认为你需要什么，另一条是广告系统认为谁愿意为你的注意力付钱。

精确定义 / 机制: 1998+ Google 的 link-based ranking 改善了 algorithmic search experience, 但早期也在寻找商业模式; 后来 Google 将 paid search ads 放在旁边, 与 algorithmic results 分开展示。Yahoo 也通过收购 Overture 和 Inktomi 组合 paid placement 与 search。2024+ 的新趋势是 AI + RAG systems, 但尚无明确最终赢家。

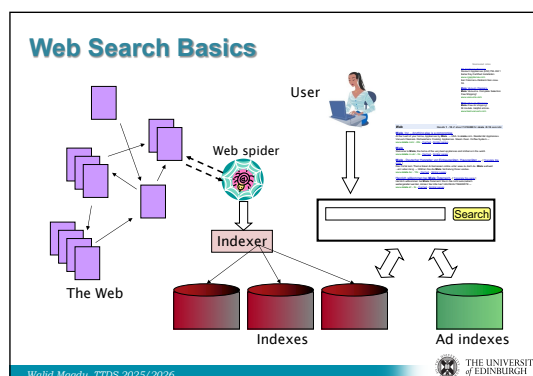
考试问法：若问 paid search 与 SEO/algorithmic ranking 区别：paid search 通过广告系统和竞价获得展示；algorithmic results 基于 relevance/quality ranking；SEO 试图影响自然结果而非直接买广告位。

3. 3. Web search basics: 从 crawler 到 indexer 再到 ranking

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11



12

6

原 PPT 第 6 页

人话直觉

搜索不是用户敲 query 后才去全网找；真正的工作早在后台完成：先派蜘蛛抓网页，整理成索引，query 来时只查索引。

精确定义 / 机制：基本组件包括 web spider/crawler、indexer、indexes/ad indexes 和 search/ranking component。Crawler 抓取网页并解析链接；indexer 处理 crawled pages，建立 searchable indexes；广告系统维护 ad indexes；用户 query 到来后，search component 在索引中检索并排序。
考试问法：组件题要按 pipeline 说清：crawl/fetch/parse/extract URLs → index pages → query-time search/ranking。别把 crawler 和 indexer 混成一个东西。

4. 4. User intent: 同一个搜索框，三种完全不同的需求

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User Need on Web Search

- **Informational** – want to learn about something (~40% / 65%)
Information Retrieval
- **Navigational** – want to go to that page (~25% / 15%)
United Airlines
- **Transactional** – want to do something (web-mediated) (~35% / 20%)
 - Access a service Seattle weather
 - Downloads Mars surface images
 - Shop Canon S410
- **Gray areas**
 - Exploratory search “see what’s there”

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13

Search Engine Optimization (SEO)

- The Trouble with Paid Search Ads:
It costs money. What’s the alternative?
- **Search Engine Optimization (SEO):**
 - “Tuning” your web page to rank highly in the algorithmic search results for selected keywords
 - Alternative to paying for placement
- Performed by companies, webmasters and consultants (“Search engine optimizers”) for their clients
- Some perfectly legitimate, some very shady

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14

7

原 PPT 第 7 页

人话直觉

搜“Information Retrieval”是在学习；搜“United Airlines”是要去官网；搜“Seattle weather”是要完成动作。结果页面应该跟着意图变。

精确定义 / 机制：Web search user needs 主要分为 informational、navigational、transactional。Informational 是想了解某事；navigational 是想去特定网站或页面；transactional 是想完成 web-mediated action，如查天气、下载图片、购物。还有 exploratory search，需求开放、边界不明确。Intent 影响 ranking features、result types 和界面呈现。

考试问法：若给 query 分类，先判断用户目标：learn、go to page、do action。若问为什么 intent 重要，答不同 intent 需要不同排序和结果展示。

5. 5. SEO simplest form: 当 tf-idf 变成考试答案模板, 网页就开始背模板

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SEO: Simplest Form

- First generation engines relied heavily on *tf/idf*
 - The top-ranked pages for the query **maui resort** were the ones containing the most **maui**'s and **resort**'s
- SEOs responded with dense repetitions of chosen terms
 - e.g., **maui resort maui resort maui resort**
 - Misleading meta-tags, excessive repetition
 - Often, the repetitions would be in the same color as the background of the web page
 - Repeated terms got indexed by crawlers
 - But not visible to humans on browsers

Pure word density cannot be trusted as an IR signal



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15

SEO word manipulating examples

- XYZ Hotel in ABC city
 - Accommodation, hotel, room, flat, travel, sights, attractions, vacation, holiday, in ABC ABC ABC
- XYZ for family advices
 - Family, couples, parents, spouse, wife, husband, fights, relationship, cheating, communication, kids, children
- XYZ Umbrellas
 - Raining, rainy, wet, weather, day

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16

8

原 PPT 第 8 页

人话直觉

如果早期引擎相信“maui resort”出现越多越相关, 网页作者就会把 maui resort 写到天荒地老, 甚至白字白底藏起来。

精确定义 / 机制: SEO 是 Search Engine Optimization, 通过调整网页使其在 algorithmic search results 中对特定 keywords 排名更高。早期 engines heavily relied on tf-idf/word density, SEOs 用 dense repetitions、misleading meta-tags、hidden text 等操纵 crawlers。pure word density 不可靠, 因为它很容易被优化成目标而不反映真实 quality/relevance。

考试问法: SEO 题要区分 white/legitimate 与 shady/spammy; 举例时写 keyword stuffing、misleading meta-tags、hidden same-color text, 并指出它们利用 term frequency/metadata signals。

6. 6. Cloaking: 给搜索蜘蛛看一张脸，给用户看另一张脸

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SEO: Cloaking

- Serve fake content to search engine spider
- Famous technique: **Black Hat**
- Kind of a spam!

Black Hat Cloaking Explained

- 1 Sites engaged in black hat SEO prepare two sets of content, one targeted for bots and the other targeted for human visitors. Both are identified by their IP address.
- 2 Since bot IP's can change, black hat informants provide a regularly updated list of bot IP addresses.
- 3 Bots are served abundant fabricated content packed with targeted keywords. This false information boosts rankings.
- 4 Human visitors often won't find the best information despite the site's high rankings.

The diagram illustrates the Black Hat Cloaking process. It shows a server with two paths: one for 'FOR HUMANS' and one for 'FOR BOTS'. A 'Human Visitor' (IP: 192.224.68.14) is shown receiving content from the human path. A 'Search Engine Bot' (IP: 208.188.255.176) is shown receiving content from the bot path. A 'LIST OF BOT IP'S' is shown with IP addresses: 208.188.255.176, 192.224.68.14, and 192.224.68.14. The diagram also shows a 'Human Visitor' and a 'Search Engine Bot' both receiving content from the server. The server is labeled 'www.blackhat.com'.

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17

Duplicate Detection

- The web is full of duplicated content
- Strict duplicate detection = exact match
 - Not as common
 - can be detected with fingerprints
- But many, many cases of **near duplicates**
 - e.g., last modified date the only difference between two copies of a page
- **Near-Duplication**: Approximate match
 - Use similarity threshold to detect near-duplicates
 - e.g., Similarity > 80% => Documents are "near duplicates"
 - Not transitive though sometimes used transitively
 - $A \approx B \ \& \ B \approx C \rightarrow$ doesn't have to mean $A \approx C$

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9

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人话直觉

cloaking 像考试时给监考老师看标准答案，转头给同学看广告传单：同一个 URL，对 crawler 和 human 端出不同内容。

精确定义 / 机制： Cloaking 是 black-hat spam technique: server 对 search engine spider 提供 fake content，对普通用户提供另一份内容。它直接破坏 index 的真实性，是比普通关键词堆砌更强的欺骗。课件紧接 duplicate detection，说明 web search 不只要找相关，还要清理重复、垃圾、伪装内容。
考试问法： 若问 cloaking，必须说 different content to spider and user；不要泛化成普通 SEO。可补充它为什么是 spam: it manipulates what gets indexed。

7. 7. Duplicate detection: 不要让同一篇文章占满索引和结果页

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The diagram illustrates the Black Hat Cloaking process. It shows a 'Human Visitor' (IP: 192.224.88.14) and a 'Search Engine Bot' (IP: 208.188.255.176). A 'List of Bot IP's' is shown, which is updated regularly. The bot is served 'abundant fabricated content packed with targeted keywords' to boost rankings, while the human visitor receives the real content. The process is controlled by 'FOR HUMANS' and 'FOR BOTS' logic.

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18

9

原 PPT 第 9 页

人话直觉

如果搜索结果前十条都是同一篇文章换了日期或镜像站，用户会觉得搜索坏了；index 也在浪费空间。

精确定义 / 机制： Web 充满 duplicate 和 near-duplicate content。Strict duplicate detection 检测 exact match，可用 fingerprints；near-duplicate detection 处理高度相似但不完全相同的 documents，例如只改 last modified date。近重复通常使用 similarity threshold，如 similarity > 80%。注意 near-duplicate relation 不一定传递： $A \approx B$ 且 $B \approx C$ 不保证 $A \approx C$ 。

考试问法：若考 duplicate detection，先区分 strict duplicate/exact match 与 near-duplicate/high similarity；再说明为什么重要：节省 index、改善 ranking 和 user experience。

8. 8. Shingles + MinHash: 把网页相似度压缩成小签名

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Duplicate Detection: MiniHash

- Features of similarity:
 - Segments of a document (natural or artificial breakpoints)
 - Shingles** (word n-grams)
 - a rose is a rose is a rose** →
a_rose_is_a
rose_is_a_rose
is_a_rose_is
a_rose_is_a
- Similarity measure between two docs (= sets of shingles)
 - Set intersection
 - Specifically (Size_of_Intersection / Size_of_Union)

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19

Shingles + Set Intersection

- Computing exact set intersection of shingles between all pairs of documents is expensive/intractable
- Approximate using a cleverly chosen subset of shingles from each (a sketch)
- Estimate $\frac{\text{size of intersection}}{\text{size of union}}$ based on a short sketch

Doc A → Shingle set A → Sketch A

Doc B → Shingle set B → Sketch B

Jaccard

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20

10

原 PPT 第 10 页

人话直觉

不要逐字比较全网每两篇文章，那像让全班每个人和每个人逐句对答案。shingles 先切成短片段，MinHash 再给每篇文档做一个小指纹。

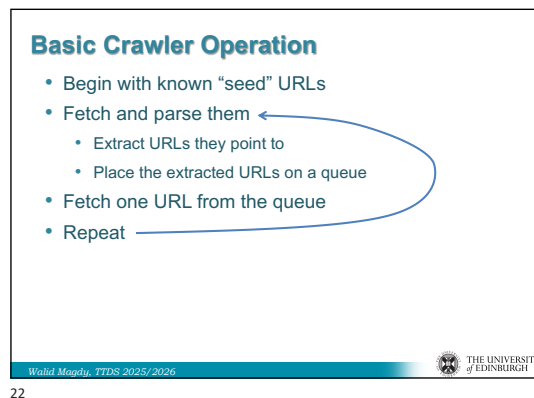
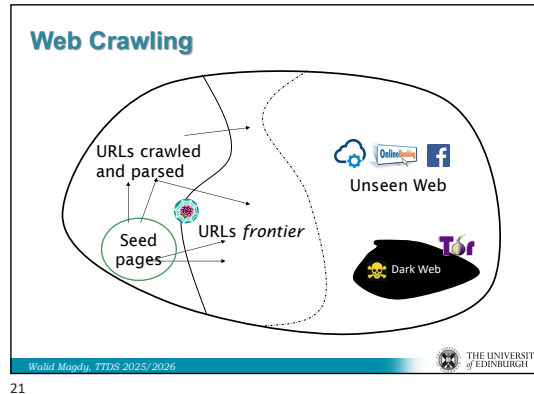
精确定义 / 机制: Shingles 是 document 的 word n-grams, 可把 document 表示为 shingle set。两个文档的集合相似度常用 Jaccard similarity, 即 intersection over union。Web-scale 下不能对所有 document pairs 精确计算 shingle intersection, 因此用 MinHash/sketch 为每个 document 生成短 signature, 近似估计 Jaccard similarity, 降低 pairwise comparison 成本。

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

考试问法: 高频题答法: document → shingles → set similarity/Jaccard → web-scale too expensive → MinHash/sketch approximation。

9. 9. Web crawling: 从 seed URLs 开始滚雪球

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11

原 PPT 第 11 页

人话直觉

crawler 像探险队：从已知城市出发，沿路牌发现新城市；但有些区域没有路牌指向，就是 dark web/unseen web。

精确定义 / 机制： Basic crawler operation: begin with known seed URLs, fetch and parse them, extract URLs they point to, place new URLs into URL frontier, repeat. URL frontier 是等待抓取的队列/集合，需要管理优先级、去重和 politeness。Dark Web 在课件图中表示 crawler 不能直接从已知 links 到达或尚未发现的部分。

考试问法： 流程题要写 seed → fetch/parse → extract links → frontier → repeat；再补 URL dedup/content dedup 和 dark/unseen web。

10. 10. Crawler architecture and frontier: 快、稳、礼貌三者拉扯

11/5/25

URL Frontier: 2 Main Considerations

- Politeness: do not hit a web server too frequently
- Priority/Freshness: crawl some pages more often than others
 - Pages whose content changes often (e.g. News sites)
- These goals may conflict each other.
 - e.g., simple priority queue fails – many links out of a page go to its own site, creating a burst of accesses to that site.
- Even if we restrict only one thread to fetch from a host, can hit it repeatedly
- Common heuristic: insert time gap between successive requests to a host that is $\gg$ time taken in most recent fetch from that host

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29

Summary

- History of Web search
- Basics of web search
- Usage of web search
- SEO
- Web crawling

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30

15

原 PPT 第 15 页

人话直觉

一个好 crawler 不是最快下载器；它像有礼貌的快递系统：不能狂敲同一家的门，还要优先送新闻这种容易过期的包裹。

精确定义 / 机制： Crawler must be polite: 尊重 robots.txt，避免过于频繁访问同一 server；must be robust: 抵抗 spider traps、malicious servers、spam/link farms；should be scalable: 支持 distributed operation；freshness 要求重新抓取已抓页面的新版本。Basic architecture 包括 DNS、fetch、parse、content seen check、URL filter、duplicate URL elimination、robots filters、URL frontier、fingerprints 等。URL frontier 的两大考虑是 politeness 与 priority/freshness，它们可能冲突：高优先级 URLs 若集中在同一 host，简单 priority queue 会造成 burst。

考试问法：若考 crawler，必须包含 politeness 和 frontier，不要只写 fetch pages。politeness 分 explicit robots.txt 和 implicit rate limiting；priority/freshness 说明哪些页面更该重抓。

考试速记版

- Early engines: keyword-based traditional IR; main issue = scalability/ranking quality.
- Sponsored search: ranking attention monetised; CPC/CPM/RPM 要分清。
- Algorithmic results $\neq$ sponsored ads; SEO $\neq$ paid search.
- User intents: informational、navigational、transactional、exploratory。
- SEO spam: keyword stuffing、misleading meta-tags、hidden text、cloaking。
- Duplicate detection: exact fingerprints vs near-duplicate threshold。
- Shingles = word n-grams; Jaccard = intersection/union。
- MinHash/sketch approximates Jaccard to avoid full pairwise comparison。
- Crawler: seed URLs $\rightarrow$ fetch/parse $\rightarrow$ extract URLs $\rightarrow$ URL frontier。
- Crawler constraints: robots.txt, implicit politeness, robustness, scalability, freshness。

一段稳妥考场回答

如果考 web crawler，可以这样答：crawler 从 seed URLs 开始，把待抓 URL 放入 URL frontier；每次取出 URL 后 fetch document、parse 页面、extract outgoing URLs，并把页面送入 indexing pipeline。系统还要做 content seen check、URL filtering、duplicate URL elimination、robots filtering 和 fingerprinting。URL frontier 不只是普通队列，它必须同时处理 priority/freshness 和 politeness：新闻等变化快或重要页面需要更频繁抓取，但 crawler 也要遵守 robots.txt 的 explicit politeness，并通过 rate limiting 避免 implicit politeness violation。一个实际 crawler 还要 robust against spider traps、malicious servers、spam/link farms，并能分布式扩展。